

Finger Detection and Counting using MediaPipe

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Project Overview

This project demonstrates real-time hand detection and finger counting using **MediaPipe** and **OpenCV** in Python. The application captures video from a webcam, detects one or two hands, identifies the 21 hand landmarks provided by MediaPipe, and counts the number of fingers that are raised.

The detected hand landmarks and the finger count are displayed live on the screen.

Features

- Real-time webcam hand detection
 - Detects up to two hands simultaneously
 - Counts raised fingers (0–5) for each hand
 - Displays hand landmarks and finger count
 - Supports both left and right hands
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Technologies Used

- Python 3.11
 - MediaPipe
 - OpenCV
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Project Structure

```
Finger_Detection_Assignment/  
|  
├─ Finger_count.py  
└─ README.md
```

Installation

1. Clone or Download the Project

Download or clone this repository to your computer.

2. Create a Virtual Environment

Windows

```
python -m venv .venv
```

3. Activate the Virtual Environment

Windows (PowerShell)

```
.\.venv\Scripts\Activate.ps1
```

Windows (Command Prompt)

```
.venv\Scripts\activate
```

4. Install the Required Libraries

```
pip install mediapipe==0.10.11 opencv-python
```

Running the Program

Run the following command:

```
python Finger_count.py
```

A webcam window will open and display:

- Hand landmarks
- Left or Right hand label
- Number of fingers raised

Press **Q** to close the application.

How It Works

1. Opens the webcam using OpenCV.
 2. Captures video frames in real time.
 3. Uses MediaPipe Hands to detect hand landmarks.
 4. Determines which fingers are raised by comparing fingertip and joint positions.
 5. Displays the finger count on the video feed.
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Expected Output

The application displays:

- Hand skeleton (21 landmarks)
- Left or Right hand label
- Finger count (0-5)

Example:

```
Right: 5  
Left: 3
```

Requirements

- Python 3.11
 - Webcam
 - MediaPipe
 - OpenCV
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Author

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Course: Robotics

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